



The Impact of Electric Vehicles on Oil Demand and Prices



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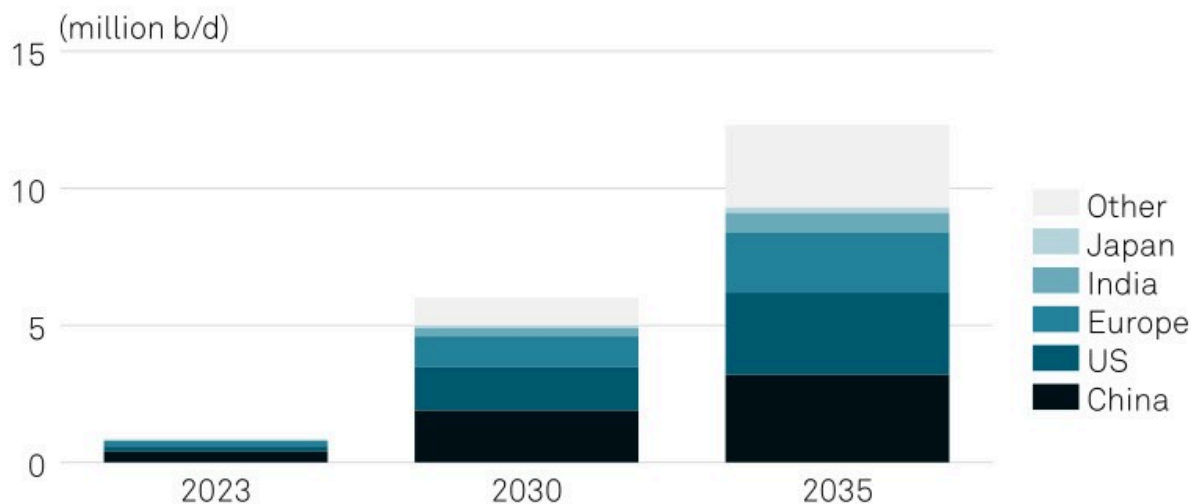
The rise of electric vehicles (EVs) is significantly reshaping the landscape of global oil demand and prices. As EV adoption accelerates, forecasts indicate a substantial displacement of oil consumption, particularly in the transportation sector, which accounts for over half of global oil demand.

Current Trends in Oil Demand

In 2022, global oil consumption reached approximately 102 million barrels per day (b/d), with nearly half attributed to road transport.

However, the impact of EVs on this demand is becoming increasingly pronounced. In 2022, EVs displaced about **1.5 million b/d** of oil, a figure projected to grow significantly in the coming years. The [International Energy Agency \(IEA\)](#) anticipates that by **2030**, this displacement could reach **6 million b/d**, and potentially **12 million b/d** by **2035** if countries adhere to their climate commitments.

Oil displacement from EVs by region



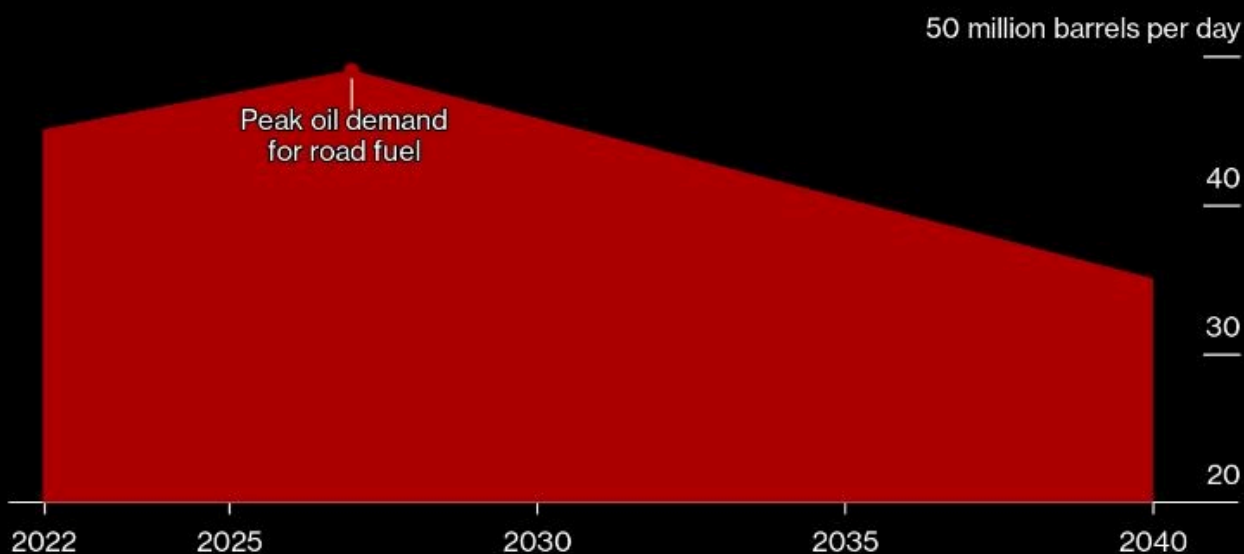
Shows oil displacement forecasts under the Announced Pledges Scenario (APS)
Source: IEAs Global EV Outlook 2024

Future Projections

Looking further ahead, [BloombergNEF](#) predicts that by **2040**, oil demand displaced by EVs could exceed **20 million b/d**. This represents a dramatic structural decline in road fuel demand, which is expected to peak at **49 million b/d** in **2027** before falling to **35 million b/d** by **2040**.

The IEA also supports this view, suggesting that peak oil demand will occur around **2030**, influenced by the growing penetration of electric vehicles in key markets like China, Europe, and North America.

Global Road Fuel Demand Peaks in 2027



Source: BloombergNEF

Note: As per BNEF's economic transition scenario, which models market-led changes.

BloombergNEF

Factors Influencing Oil Prices

Despite the anticipated decline in oil demand due to EVs, the relationship between oil demand reduction and oil prices is complex. The decline in consumption does not automatically lead to lower prices; rather, it could result in price volatility depending on supply dynamics. If new oil supply investments decrease faster than demand diminishes, prices may remain high or unstable.

Additionally, geopolitical factors and [OPEC](#) 's strategies to manage production levels can also impact pricing trends.

Structural Changes in the Oil Market

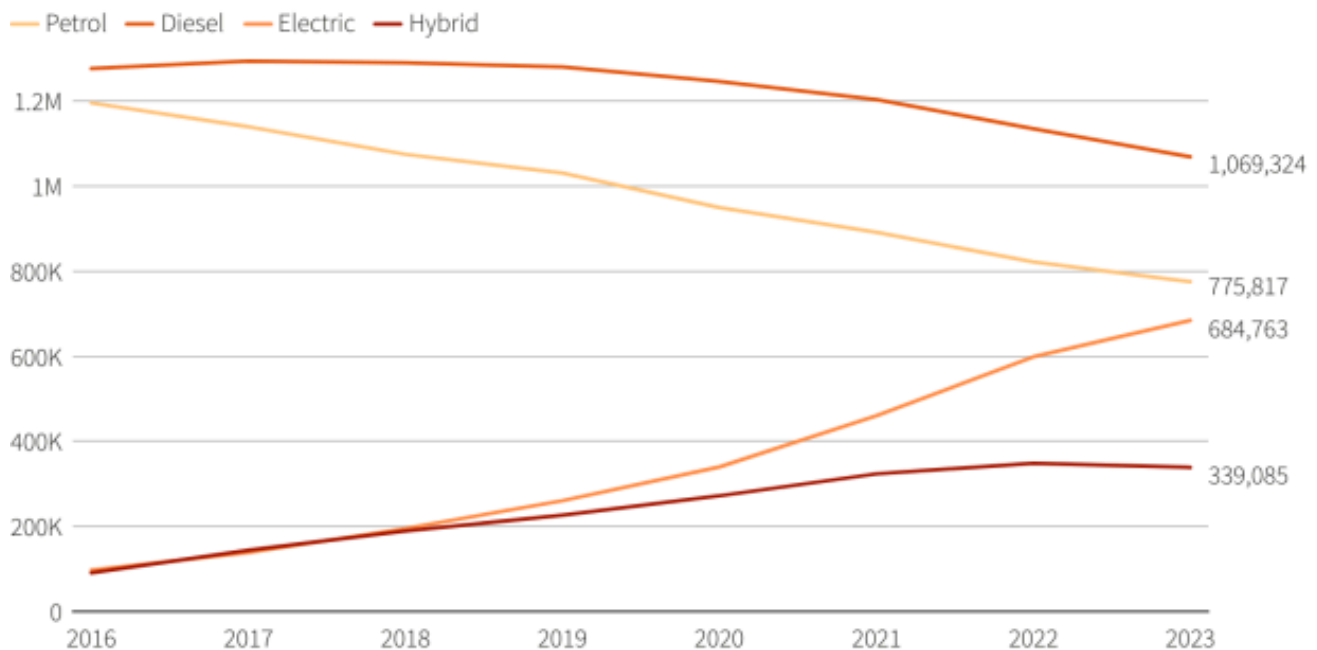
The shift towards electrification is prompting significant adjustments within the oil industry. Major oil companies are beginning to prepare for a future with reduced oil volumes as they adapt their business models to align with net-zero emissions goals. European firms are reportedly leading this transition, with companies like Equinor and BP taking proactive steps toward sustainable energy initiatives.

Case Studies on EV Impact

Case Study 1: Norway's EV Adoption and Its Impact on Oil Demand

Norway leads the world in EV adoption, with over 80% of new cars sold in 2022 being electric. As a result, the demand for gasoline and diesel in Norway has significantly declined. This shift demonstrates how strong government incentives, public awareness, and extensive charging infrastructure can drive rapid changes in oil consumption patterns.

Electric cars could overtake petrol cars in Norway by end-2024



Note: majority of hybrid cars are plug-in hybrids with petrol engines

Source: Statistics Norway (SSB); the Norwegian Public Roads Administration

Case Study 2: China's EV Market Boom and Global Oil Prices

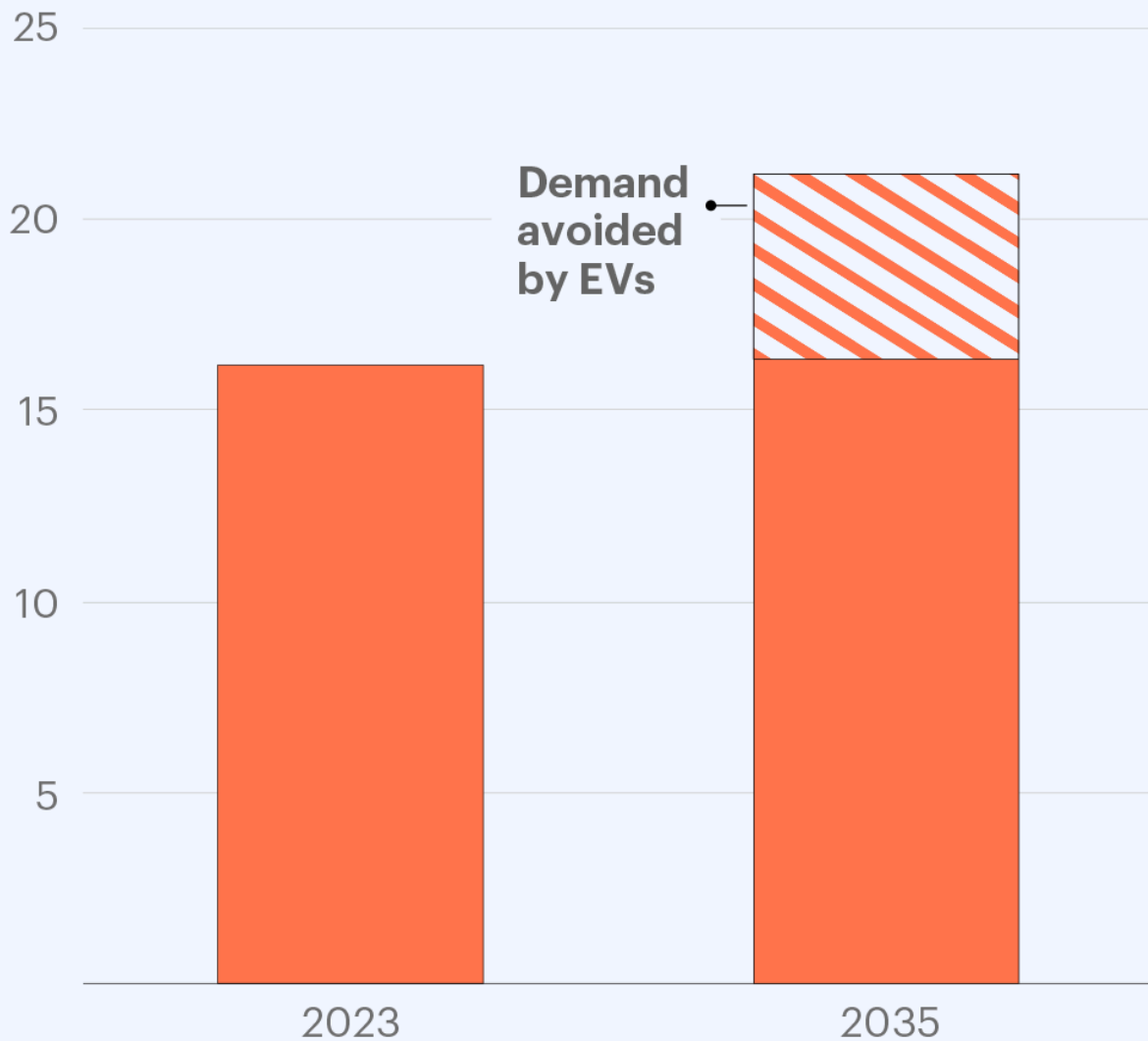
China, the world's largest automotive market, has aggressively promoted EV adoption to reduce pollution and meet climate goals. In 2023, China's EV sales represented nearly 30% of the global EV market. This shift has started impacting global oil demand, as China is a major consumer. With continued growth, China's EV adoption could exert more pressure on global oil prices in the coming years.

EVs are set to cap China's oil demand growth



Oil demand in China under today's policy settings, 2023-2035

mb/d



Case Study 3: California's Push for EVs and Effects on U.S. Oil Consumption

California, a trendsetter in environmental regulations, has set ambitious targets to phase out gasoline vehicles by 2035. With one of the highest EV adoption rates in the U.S., California's policies are already reducing gasoline consumption and influencing neighboring states. This

reduction in oil demand within a major U.S. economy could accelerate national trends toward reduced oil dependency.

Case Study 4: Saudi Arabia's Economic Diversification in Response to EV Growth

Saudi Arabia, one of the world's largest oil producers, is proactively responding to declining oil demand due to EV growth. Through its Vision 2030 initiative, Saudi Arabia is investing in renewable energy and developing industries outside of oil, including tourism and tech. This diversification strategy reflects the country's need to adapt to a changing global energy market.



Conclusion

In summary, the impact of electric vehicles on oil demand is poised to be transformative. As EV adoption accelerates, particularly supported by favorable policies and technological advancements in battery efficiency, a substantial portion of current oil consumption will be displaced. While this trend presents challenges for traditional oil markets, it also offers opportunities for innovation and investment in alternative energy sources. The coming decade will be crucial as these dynamics unfold and reshape the global energy landscape.

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